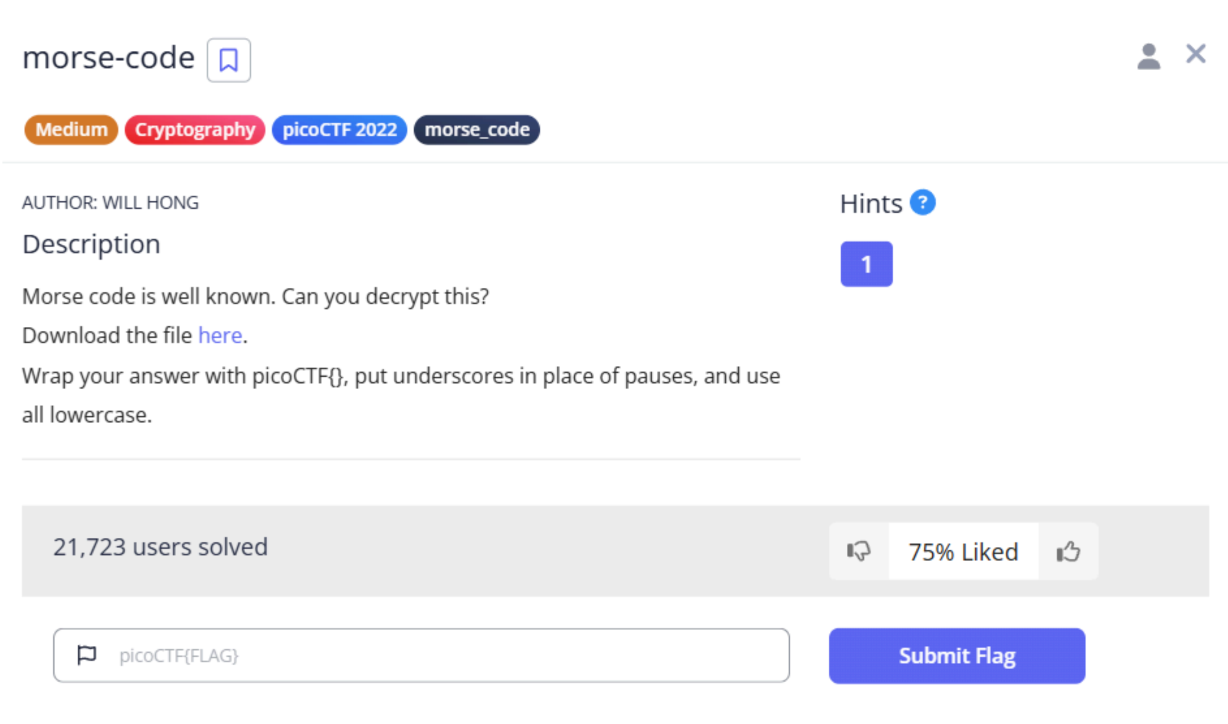
HERNANDO, PANTINE B. **BSIT-3**
CHALLENGE - 12 - CRYPTOGRAPHY



Challenge Overview:

The challenge provides an audio file (typically a `.wav` or `.mp3`) containing Morse code beeps. To solve it, you need to decode the audio, format it according to specific rules, and wrap it in the flag format.

1. Analysis

The description gives us three critical formatting rules for the final flag:

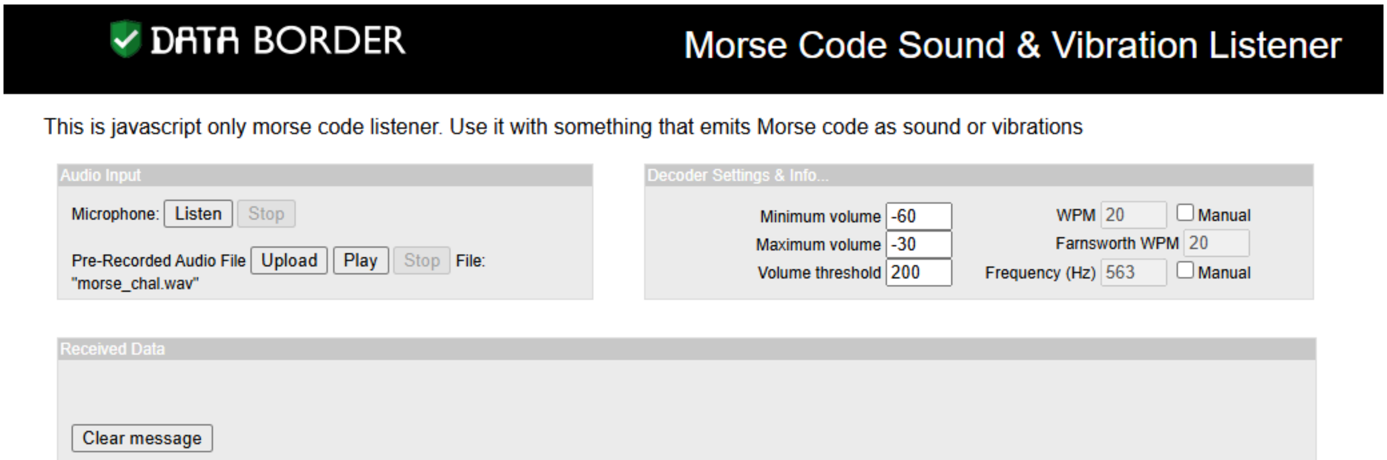
1. **Wrap** the answer with `picoCTF{}`.
2. **Replace pauses** (gaps between words) with underscores `_`.
3. Use **all lowercase** letters.

2. The Solution Path

Method A: Automated Decoding (Easiest)

Since human ears can be fallible (and slow), the most efficient way to solve this is using an online spectrum analyzer or Morse decoder.

1. **Download** the audio file from the challenge link.
2. **Upload** it into an online tool like Data Border https://databorder.com/transfer/morse-sound-receiver/ The tool will process the audio beeps and output the text automatically.

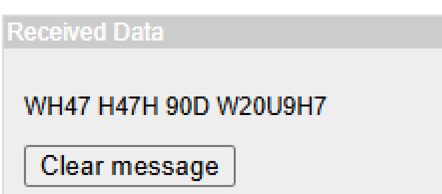


3. Formatting the Flag

Let's say the decoded Morse code results in: `WH47 H47H G0D WR0U6H7`

Following the challenge instructions:

- **Lowercase:** `wh47 h47h g0d wr0u6h7`
- **Underscores for pauses:** `wh47_h47h_g0d_wr0u6h7`
- **Wrap in picoCTF{}:** `picoCTF{wh47_h47h_g0d_wr0u6h7}`



4. Key Takeaways

- **Signal Processing:** In CTFs, audio "cryptography" is often just data encoded in a different physical medium (sound).
- **Instruction Adherence:** Most "incorrect" submissions on this challenge come from forgetting the lowercase rule or using spaces instead of underscores.